Clinical Guideline for:
Critical Care Management of Acute Asthma in Adult Patients

SUMMARY

This guideline outlines the assessment and treatment steps for an adult patient who has been referred for assessment by the intensive care team.

It can also be used as a guide to help management of ICU patients with severe or life-threatening asthma.

KEY POINTS

- The primary pathophysiological problem in life-threatening asthma is expiratory gas-flow limitation.
- Wheezing does not correlate well with the degree of airway obstruction.
- Assess the severity of the acute asthma attack and use a step-wise approach to treatment.
- Always be wary of normocapnia and consider it in clinical context; it may be a sign of patient deterioration or exhaustion.
- Initiation of invasive ventilation should ALWAYS be discussed with the duty ICU consultant - who ideally should be present at intubation. If time does not permit, the most senior clinician available should be present.

NB. This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

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1. ASSESSMENT OF A PATIENT WITH ACUTE ASTHMA

To classify the severity of asthma use a combination of:

- Clinical features (e.g. wheeze, breathlessness, tachypnoea, tachycardia, cyanosis, silent chest).
- PEF or FEV_1, if practical.
- Pulse oximetry.
- ABG.
- CXR (in patients with life-threatening asthma, if pneumothorax or consolidation suspected or if the patient requires IPPV.)

Remember:

1. **Not all wheezing is asthma**. Other causes of wheezing include: pulmonary oedema, COPD, pneumonia, anaphylaxis, foreign bodies, pulmonary embolism, and subglottic mass.

2. Degree of wheeze does not correlate well with the degree of airway obstruction and absence of wheezing may indicate critical airway obstruction.

Classification of severity of acute asthma*:*

Moderate asthma:	Acute Severe asthma:
PEF >50–75% best or predicted No features of acute severe asthma	Any one of: • PEF 33–50% best or predicted • respiratory rate ≥25/min • heart rate ≥110/min • inability to complete sentences in one breath
Life-threatening asthma:	**Near-fatal asthma:**
In a patient with severe asthma, any one of: • PEF <33% best or predicted • SpO2 <92% • PaO2 <8 kPa • normal PaCO2 (4.6–6.0 kPa) • silent chest • cyanosis • poor respiratory effort • arrhythmia • exhaustion • altered conscious level • hypotension	Raised PaCO2 and/or requiring mechanical ventilation with raised inflation pressures

Note: *Adapted from British Thoracic Society/SIGN 153 –British Guideline on the Management of Asthma. 2016*

2. TREATMENT OF A PATIENT WITH ACUTE ASTHMA

Immediate treatment of acute asthma

- Oxygen via non-rebreather mask- aiming for sats of >94%.
- Salbutamol 5mg via an oxygen-driven nebuliser.
- Ipratropium bromide 500mcg via an oxygen driven nebuliser and continue qds.
- Prednisolone 40-50mg orally or hydrocortisone 100-200mg IV
- Antibiotics not usually required – consider in the presence of infective symptoms, consolidation on CXR, high inflammatory markers.
- No sedatives of any kind.

IF LIFE THREATENING FEATURES ARE PRESENT:

- These critically ill patients should have continuous close monitoring (HR, ECG, RR, BP, SpO2 as a minimum).
- Add IV Magnesium sulphate 2g infusion over 20 min (unless already given) (cardiac monitoring required).
- Give nebulised salbutamol more frequently, up to continuously if required.

Subsequent Treatment

IF PATIENT NOT IMPROVING AFTER 15-30 MIN:

- Give nebulised salbutamol continuously.
- Consider use of an IV salbutamol only if inhaled therapy cannot be used reliably, such as if patient is coughing excessively, response to repeated nebulisers is poor or patient becomes moribund
    - Give IV salbutamol 250µg IV slowly over 20 min, followed by an infusion at 5-20 µg/min. (lactic acidosis, hypokalaemia and arrhythmias may develop with salbutamol. Monitor potassium and replace if necessary).
- IV Magnesium sulphate 2g infusion over 20 min (unless already given).
- Patients are often dehydrated or hypovolemic and are likely to benefit from IV fluids.

IF PATIENT STILL NOT IMPROVING:

- **Further management should be in close conjunction with the ICU consultant from here**
- Some patients with near fatal or life-threatening asthma with poor response to initial treatment may gain additional benefit from the addition of IV aminophylline.
- There is limited and inconclusive evidence for the use of NIV in severe asthma and is not recommended.
- Consider the potential need for intubation if worsening or persisting hypoxaemia or hypercapnia, exhaustion, confusion, drowsiness, coma or respiratory arrest.
- After initial ventilation settings additional therapies which can be considered (in conjunction with the ICU consultant) include ketamine infusion, adrenaline infusion and inhalational anaesthetics.

3. MECHANICAL VENTILATION

Patients requiring intubation and ventilation

If considering intubating a patient with acute asthma, inform the duty ICU consultant. Ideally the consultant should be present at intubation, or if time does not permit, the most senior clinician available.

Intubation can be life-saving, but has a higher incidence of complications in acute severe asthma relative to other causes of respiratory failure and can be hazardous.

Absolute indications for intubation are coma, respiratory or cardiac arrest and severe refractory hypoxaemia.

Relative indications include an adverse trajectory of response to initial management, fatigue and somnolence.

Induction

Aim for a size 9.0 mm endotracheal tube in males and 8mm in females if possible
Induction Drugs:

- Ketamine may be useful for the bronchodilating effect
- Rocuronium or Suxamethonium

Consider causes of hypotension peri-intubation: hypovolaemia, pneumothorax, dynamic hyperinflation (breath-stacking), loss of endogenous catecholamines

Initial ventilation settings

- The initial goals of mechanical ventilation are to correct hypoxaemia and to buy time for medical management to work, whilst avoiding complications such as barotrauma and haemodynamic compromise

- Initial ventilator settings should adopt relatively low rates (8-12 breaths min^{-1}), tidal volumes of 4–6 ml kg-1, FiO_2 to maintain oxygen saturations >92%, relatively long expiratory times (I:E > 1:3) and no PEEP.

(Once the patient is on the ICU, the degree of gas trapping (intrinsic PEEP) can be assessed using the ventilator to see whether additional PEEP may be of benefit).

- If using volume controlled ventilation, appropriate goals would be to achieve a Pplat <35 cm H2O with pH > 7.2:

4. DRUGS: QUICK REFERENCE GUIDE

Adrenaline	infusion rate 0.01-0.1µg/kg/min
Alfentanil	infusion rate 0.5-1 µg/kg/min
Aminophylline	5mg/kg IV loading dose over 20min (omit loading dose if patient already on oral therapy) – then infusion of 0.5-0.7mg/kg/hr
Hydrocortisone	100-200mg IV starting dose – then 100mg IV 6-hourly
Ipratropium	500µg nebulised 4-6 hourly
Ketamine	induction dose 1-2mg/kg IV. Infusion rate 10-30 µg/kg/min
Magnesium Sulphate	1.2 -2g IV over 20min
Prednisolone	40-50mg orally starting dose and continue daily
Propofol	infusion rate for sedation 1-4mg/kg/hour
Rocuronium	Induction dose 1-1.2mg/kg**.** Infusion rate 300-600µg/kg/hr
Salbutamol (nebulised)	5mg via oxygen driven nebuliser – can be given back-to-back if needed
Salbutamol (IV)	IV salbutamol 250µg IV slowly over 20min, then infusion at 5-20 µg/min
Suxamethonium	1.5mg/kg

5. REFERENCES

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6. PUBLICATION DETAILS

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